

Report: March

Rishi Vora

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1. SHORT SUMMARY OF PREVIOUS WORK

In February, I focused on making the `Julia` QDDPM implementation fast. I fixed major Zygote/Yao autodiff failures, switched to a Wasserstein objective with explicit Zygote gradients using the envelope theorem, tried out several optimizers (gradient-based and gradient-free).

2. MY WORK

2.1. BENCHMARKING OPTIMIZERS

After a successful implementation of gradient-based optimization using the envelope theorem, I benchmarked several optimizers on a 4-qubit system. The optimizers included:

- Adam (baseline)
- RAdam
- NAdam
- AdamW
- AdaBelief
- Lion
- AMSGrad

The results showed that AdaBelief and AMSGrad performed quite better than the baseline Adam optimizer.

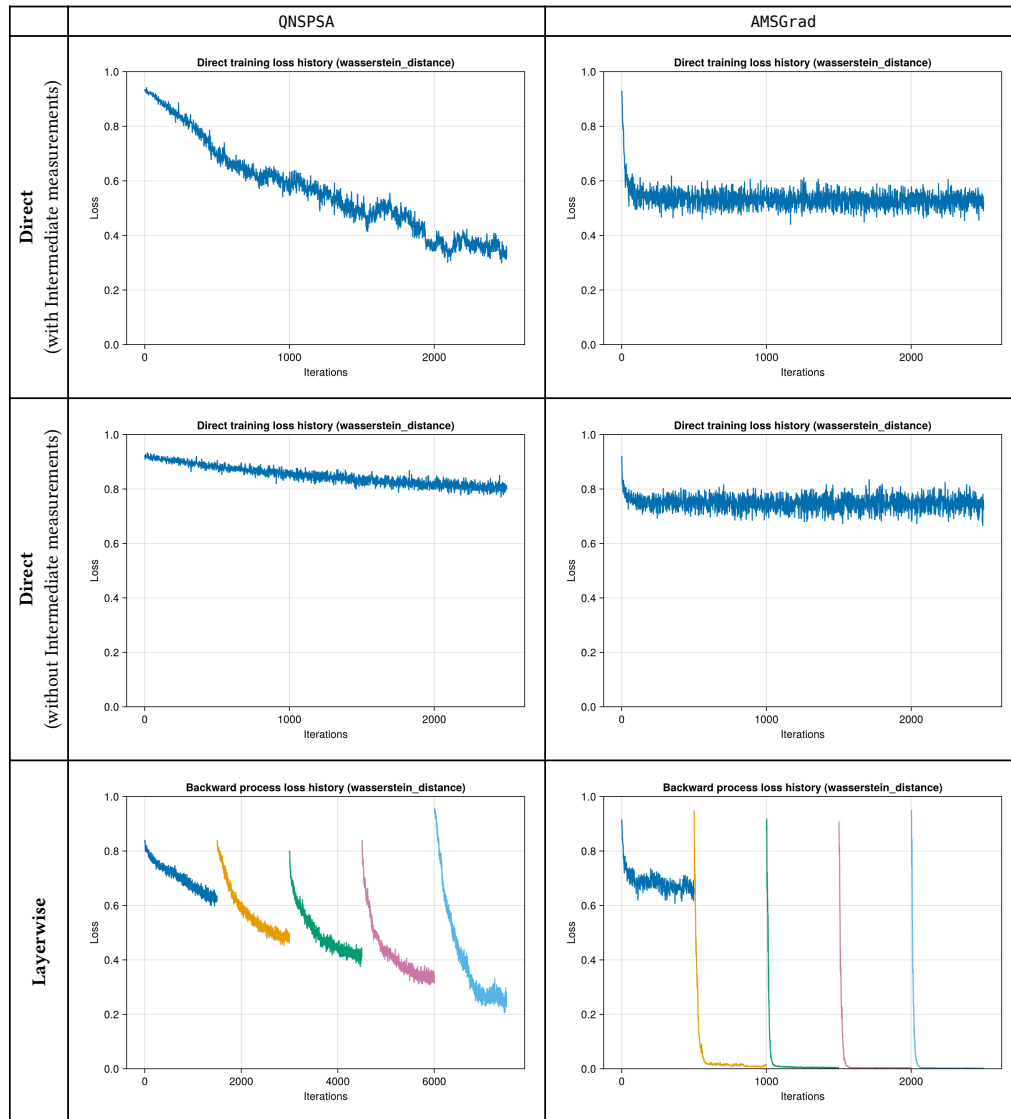
Taking these two optimizers forward, I benchmarked them on a 6-qubit system. AMSGrad performed better and was hence chosen as the optimizer for training larger systems.

2.2. COMPARISON WITH QUdT

I compared the performance of my implementation with Quantum Direct Transport (QuDT) model as per described in [1]. The basic idea to train all the parameters to minimize the distance to the target ensemble. Out of curiosity, I also tweaked the QuDT to include intermediate measurements on ancilla, with the intuition that it would help dump the entropy of the system to make the convergence better.

Parameters

```
n_qubits: 4
n_ancilla: 2
n_layers: 60 (Total)
```



It certainly performed better than the vanilla QuDT, but it was still not good enough. Curiously, the gradient-less QNPSA optimizer performed better than the gradient-based AMSGrad optimizer. But it still no match to AMSGrad with step-wise training. With 6-qubits, the gap widens even more; AMSGrad converged far quicker and better than the QuDT-based approaches.

Hence, the conclusion is that the step-wise training approach with a Wasserstein objective and explicit gradients has a much better chance of working for larger as well as chaotic systems.

2.3. KICKED ROTATOR

I have started reading up the theory of the kicked rotator and the Quantum kicked rotator.

3. PLANNED WORK

To understand the theory of the kicked rotator. Also to do preliminary testing of 10-qubit systems with the AMSGrad optimizer. Then try to generate Quantum kicked rotator states with it.

BIBLIOGRAPHY

- [1] B. Zhang, P. Xu, X. Chen, and Q. Zhuang, “Generative quantum machine learning via denoising diffusion probabilistic models,” *Phys. Rev. Lett.*, vol. 132, no. 10, p. 100602, Mar. 2024.